

# YUZE LI

Email: [yuzeli0120@gmail.com](mailto:yuzeli0120@gmail.com) | Tel: +1 (203)-589-5687

Homepage: <https://yuzeli-0120.github.io>

## EDUCATION

**College of Optical Science and Engineering, Zhejiang University**

*B.S., Optical Engineering*

**Zhejiang, China**

*Expected June 2026*

- GPA: 4.10/4.3
- Rank: 3/97
- Math & PE: All mathematical and physical courses have high scores.

## RESEARCH EXPERIENCE

**Researches concerning nonlinear optics and quantum optics.**

**Yale University**

Advisor: Prof. Hong Tang (Dept. of Electrical & Computer Engineering)

Jul. 2025 - Now

- Doing research work concerning nonlinear and quantum optics in Prof. Tang's group.
- Received a series of training on AlN chip calibration and measurement. About to learn how to fabricate and calibrate periodic poled crystals (by SHG method etc.).

**Mode field diameter controlling with laser direct writing method.**

**Zhejiang University**

Advisor: Prof. Jianrong Qiu (Dept. of Optical Science and Engineering)

Sept. 2024 – Jun. 2025

- I have experimentally verified that the mode field diameter of the waveguide can indeed exhibit regular variations based on this processing method. Further research will continue to follow up on this.
- Yuze Li, Jianrong Qiu et al. [Tapered three-dimensional waveguide with controllable mode field diameter by composite laser direct writing method.](#) (A Poster presented at 6<sup>th</sup> Ultrafast Laser Precision Processing Technology and Application Symposium)

**Medical Image Restoration Based on Implicit Neural Representations**

**Zhejiang University**

Advisor: Prof. Huafeng Liu (Dept. of Optical Science and Engineering)

Nov. 2024 - Now

- The primary objective of this project is to leverage the robustness of Implicit Neural Representations (INR) to Gaussian noise as a novel approach for enhancing medical images. By integrating INR with the Neighbor2Neighbor network, we aim to restore medical images heavily affected by noise. The improved model has already achieved promising results, and I am currently preparing a paper as the first author.

**Micro-nano fiber tapering system**

**Zhejiang University**

Advisor: Dr. Yaoguang Ma (Dept. of Optical Science and Engineering)

Nov. 2023 - Jun. 2024

- The main focus of the experiment was on fiber morphology fitting, heating zone calibration, and the fabrication of long-waist MNFs (micro-nano fibers) with a diameter at the micrometer scale using a vertical electrically heated MNF tapering system. I also developed a custom Python algorithm to automate the measurement of optical fiber diameters under a microscope, achieving a measurement accuracy of over 95%.

**Spectral Food Inspection — Non-contact Intelligent Freshness System for Food**

**Zhejiang University**

Advisor: Prof. Sailing He (Dept. of Optical Science and Engineering)

Mar. 2024 - Aug. 2024

- Using hyperspectral imaging to non-destructively and rapidly assess food freshness, I conducted multiple measurements and calibrations to determine the most suitable light source parameters for different fresh products.

**Python-based Mouse Scroll Wheel Defect Detection Project**

**Zhejiang University**

Advisor: Prof. Kaiwei Wang (Dept. of Optical Science and Engineering)

- In this project, python was used to perform a visual recognition task utilizing the Canny operator and gradient threshold detection algorithm, demonstrating high robustness and achieving 100% accuracy.
- Open source: <https://github.com/YuzeLi-0120/Python-based-Mouse-Scroll-Wheel-Defect-Detection-Project>

## HONORS

- |                                                                                                     |           |
|-----------------------------------------------------------------------------------------------------|-----------|
| • First Prize in the National College Students' Mathematical Contest, Zhejiang Division.            | 2023      |
| • External Scholarship, Yongping Scholarship (Student representative in Duan's Homecoming Meeting.) | 2023-2024 |
| • University-level Second Class Scholarship                                                         | 2022-2023 |
| • Third Prize in the Zhejiang Province Physics Innovation Competition.                              | 2023      |

## EXTRACURRICULAR ACTIVITIES & INTERESTS

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|-------------------------------------------------------------------------------------------------|-----------------------|
| • Engaged as a team leader in public legal education activities organized by the Procuratorate. | Jul. 2024 - Aug. 2024 |
| • Volunteer Service at Campus Cainiao Stations.                                                 | Nov. 2022-May. 2023   |
| • Volunteer Service at the Global Artificial Intelligence Conference.                           | Nov. 2022             |

## SKILLS

**Programming:** C, Python, MATLAB

**Experimental Skills:** Measurement and calibration of ring resonators, waveguides and fibers. Platform controlling.

**English:** TOEFL *100* out of 120. (Reading 25, Listening 30, Speaking 22, Writing 23)